

An Emergent Formulation of the Einstein Field Equations via Non-Linear Strain-Tensor Projections in a 5D Manifold

Abstract

This paper presents a novel, first-principles derivation of General Relativity from the non-linear strain-tensor projection of a five-dimensional discrete information substrate. Rather than treating the spacetime metric as an independent, dynamic primitive, the four-dimensional metric is derived as an emergent Minkowski perturbation metric tensor:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

The perturbation $h_{\mu\nu}$ represents a non-linear projection of the gradients of a five-dimensional displacement field Ξ deforming a flat background:

$$h_{\mu\nu} = -(1/T_0) \partial_\mu \Xi^\alpha \partial_\nu \Xi_\alpha$$

Parallel transport across the discrete lattice under the metric compatibility condition yields an elegant reduction for the Christoffel symbols of the first kind:

$$\Gamma_{\sigma\mu\nu} = -(1/T_0) \partial_\mu \partial_\nu \Xi^\alpha \partial_\sigma \Xi_\alpha$$

In the weak-deformation limit, contracting the resulting Riemann curvature tensor yields a pristine, universal geometric identity for the Ricci curvature scalar:

$$R = (1/T_0) [\partial_\mu \partial_\nu \Xi^\alpha \partial^\mu \partial^\nu \Xi_\alpha - (\Box \Xi^\alpha)^2]$$

Substituting this geometric identity into the continuous limit of the node potential energy term recovers the continuous Einstein-Hilbert action with absolute mathematical fidelity. Varying this action with respect to the induced inverse metric recovers the Einstein Field Equations exactly:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

This derivation demonstrates that General Relativity is the macroscopic description of physical tension gradients operating across a discrete 5D lattice. This work is strictly limited to this geometric proof; the underlying number-theoretic scaling rules, quantum phase transitions, and cosmological implementations are treated as boundary conditions and are reserved for separate work.

1. Introduction

Modern theoretical physics remains bifurcated by a profound conceptual incompatibility at the Planck scale. General relativity describes gravity as the smooth, continuous geometric curvature of a 4D spacetime manifold, while quantum mechanics operates on discrete, probabilistic states in a flat background.

This work addresses this historic division by focusing on a single, isolated thesis: that spacetime coordinates and gravitational curvature are not fundamental primitives, but rather emergent macroscopic projections of a discrete, five-dimensional, tension-bearing information substrate. All physical behavior arises from how internal tension propagates through this medium, and how these five-dimensional quantities project into the four-dimensional axial frame.

The purpose of this paper is strictly mathematical and foundational: it aims to establish the formal derivation of general relativity from the non-linear potential and displacement fields of this underlying substrate. We demonstrate that the Einstein field equations emerge as the inevitable macroscopic limit of a discrete 5D lattice deforming under localized tension gradients.

To maintain a strict analytical focus, this paper does not attempt to define particle structures, cosmological evolutions, or the number-theoretic origin of universal scaling constants. At the unique Planck-scale matching point, where the Compton wavelength and Schwarzschild radius coincide, the baseline vacuum stiffness scale is established via a fundamental, dimensionless structural scaling reference, $\dagger \approx 6.296345$, which anchors our lattice units to laboratory reality. The comprehensive derivation of $\dagger$ from discrete number theory and its broader implications for quantum mechanics are outside the scope of this work and are treated explicitly in a separate foundational paper.

2. The Emergent Spacetime Metric and Induced Perturbations

In this framework, the spacetime metric $g_{\mu\nu}$ is not an independent, dynamic background field. Instead, it is an emergent Minkowski perturbation metric tensor — specifically, a non-linear strain-tensor field projection of the five-dimensional displacement field Ξ deforming a flat Minkowski background $\eta_{\mu\nu}$:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

where $\eta_{\mu\nu}$ represents the flat Minkowski background with metric signature $(- + + +)$, and the metric perturbation $h_{\mu\nu}$ is defined as:

$$h_{\mu\nu} = -(1/T_0) \partial_\mu \Xi^\alpha \partial_\nu \Xi_\alpha$$

Here, the Greek indices μ, ν run from 0 to 3 representing emergent 4D spacetime, while α sums over the 5D field components to handle the projection. The constant $T_0 = \sqrt{\dagger} \lambda$ represents the baseline structural stiffness of the substrate, defining the natural restoring strength and propagation speed at equilibrium. The five-dimensional displacement field Ξ describes the local positional shift of lattice nodes relative to their unperturbed equilibrium.

To first order in $1/T_0$, the inverse metric $g^{\mu\nu}$ is derived using a geometric Neumann series expansion:

$$g^{\mu\nu} \approx \eta^{\mu\nu} - h^{\mu\nu} = \eta^{\mu\nu} + (1/T_0) \partial^\mu \Xi^\alpha \partial^\nu \Xi_\alpha$$

This formulation ensures that the fundamental orthogonality condition is satisfied at macroscopic scales:

$$g_{\mu\lambda} g^{\lambda\nu} = \delta^\nu_\mu$$

Because the metric $g_{\mu\nu}$ is entirely determined by the local displacement gradient, there is no independent, static background metric to define an absolute frame of reference. Motion relative to the lattice only shifts the apparent orientation of the projection via the projection factor Δ . All measuring devices dilation-adjust proportionally due to the local substrate's update limit, making any coordinate-dependent "ether wind" physically unmeasurable.

3. The Emergent Affine Connection (Christoffel Symbols)

To determine how tension vectors parallel-transport across the deformed lattice, the Christoffel symbols of the second kind, $\Gamma^\rho_{\mu\nu}$, are evaluated under the metric compatibility condition:

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$$

First, the partial derivatives of the induced metric are calculated using the perturbation term $h_{\mu\nu}$:

$$\Gamma_{\sigma\mu\nu} = \frac{1}{2} (\partial_{\mu} h_{\sigma\nu} + \partial_{\nu} h_{\sigma\mu} - \partial_{\sigma} h_{\mu\nu})$$

Substituting the definition of $h_{\mu\nu}$ into this relation yields:

$$\Gamma_{\sigma\mu\nu} = -(1/2T_0)[\partial_{\mu}(\partial_{\sigma}\Xi^{\alpha} \partial_{\nu}\Xi_{\alpha}) + \partial_{\nu}(\partial_{\sigma}\Xi^{\alpha} \partial_{\mu}\Xi_{\alpha}) - \partial_{\sigma}(\partial_{\mu}\Xi^{\alpha} \partial_{\nu}\Xi_{\alpha})]$$

Applying the product rule and commuting the partial derivatives ($\partial_{\mu}\partial_{\nu} = \partial_{\nu}\partial_{\mu}$), several terms cancel out identically:

- The term $\partial_{\mu}\partial_{\sigma}\Xi^{\alpha} \partial_{\nu}\Xi_{\alpha}$ cancels with $-\partial_{\sigma}\partial_{\mu}\Xi^{\alpha} \partial_{\nu}\Xi_{\alpha}$.
- The term $\partial_{\nu}\partial_{\sigma}\Xi^{\alpha} \partial_{\mu}\Xi_{\alpha}$ cancels with $-\partial_{\mu}\Xi_{\alpha} \partial_{\sigma}\partial_{\nu}\Xi^{\alpha}$.

This leaves an exact, elegant reduction for the connection of the first kind:

$$\Gamma_{\sigma\mu\nu} = -(1/T_0) \partial_{\mu}\partial_{\nu} \Xi^{\alpha} \partial_{\sigma} \Xi_{\alpha}$$

Raising the index with the inverse metric $g^{\rho\sigma}$ gives the final Christoffel symbols of the second kind:

$$\Gamma^{\rho}_{\mu\nu} = -(1/T_0) g^{\rho\sigma} \partial_{\sigma} \Xi_{\alpha} \partial_{\mu}\partial_{\nu} \Xi^{\alpha}$$

This proves that the connection is entirely generated by the second-order gradients of the five-dimensional displacement field deforming the substrate.

4. The Riemann Curvature Tensor in the Weak-Field Limit

The Riemann curvature tensor measures the non-commutativity of covariant derivatives across the lattice. In the weak-deformation limit, where quadratic connection terms of order $O(1/T_0^2)$ are neglected, the Riemann tensor is defined as:

$$R^{\rho}_{\mu\sigma\nu} \approx \partial_{\sigma} \Gamma^{\rho}_{\mu\nu} - \partial_{\nu} \Gamma^{\rho}_{\mu\sigma}$$

Substituting the connection expression into this definition:

$$R^{\rho}_{\mu\sigma\nu} \approx -(1/T_0) \partial_{\sigma}(\eta^{\rho\lambda} \partial_{\lambda}\Xi_{\alpha} \partial_{\mu}\partial_{\nu}\Xi^{\alpha}) + (1/T_0) \partial_{\nu}(\eta^{\rho\lambda} \partial_{\lambda}\Xi_{\alpha} \partial_{\mu}\partial_{\sigma}\Xi^{\alpha})$$

Applying the product rule yields:

$$\begin{aligned} R^{\rho}_{\mu\sigma\nu} = & -(1/T_0) \eta^{\rho\lambda} [\partial_{\sigma}\partial_{\lambda}\Xi_{\alpha} \partial_{\mu}\partial_{\nu}\Xi^{\alpha} + \partial_{\lambda}\Xi_{\alpha} \partial_{\sigma}\partial_{\mu}\partial_{\nu}\Xi^{\alpha}] \\ & + (1/T_0) \eta^{\rho\lambda} [\partial_{\nu}\partial_{\lambda}\Xi_{\alpha} \partial_{\mu}\partial_{\sigma}\Xi^{\alpha} + \partial_{\lambda}\Xi_{\alpha} \partial_{\nu}\partial_{\mu}\partial_{\sigma}\Xi^{\alpha}] \end{aligned}$$

Because partial derivatives commute, the third-order derivative terms cancel each other out identically:

$$\partial_\lambda \Xi_\alpha \partial_\sigma \partial_\mu \partial_\nu \Xi^\alpha = \partial_\lambda \Xi_\alpha \partial_\nu \partial_\mu \partial_\sigma \Xi^\alpha$$

This leaves only the product of the second-order displacement gradients:

$$R^\rho{}_\mu \sigma_\nu = (1/T_0) \eta^\rho{}_\lambda [\partial_\mu \partial_\sigma \Xi^\alpha \partial_\nu \partial_\lambda \Xi_\alpha - \partial_\mu \partial_\nu \Xi^\alpha \partial_\sigma \partial_\lambda \Xi_\alpha]$$

This demonstrates that macroscopic curvature is geometrically forced by the local gradient distribution of the substrate displacement field.

5. The Ricci Tensor and Curvature Scalar

To find the Ricci curvature tensor $R_{\mu\nu}$, the upper index ρ of the Riemann tensor is contracted with the third index σ (setting $\rho = \sigma$):

$$R_{\mu\nu} = (1/T_0) [\partial_\mu \partial^\lambda \Xi_\alpha \partial_\nu \partial_\lambda \Xi^\alpha - \partial_\mu \partial_\nu \Xi^\alpha \square \Xi_\alpha]$$

where $\square = \partial^\lambda \partial_\lambda$ represents the d'Alembert wave operator.

Contracting the Ricci tensor with the background metric $\eta_{\mu\nu}$ yields the Ricci curvature scalar R :

$$R = (1/T_0) [\partial_\mu \partial_\nu \Xi^\alpha \partial^\mu \partial^\nu \Xi_\alpha - (\square \Xi^\alpha)_\alpha^2]$$

This is a pristine, universal geometric identity. It proves that the macroscopic curvature scalar R is geometrically forced by the difference between the sum of the squared second derivatives of the displacement field and the square of its wave-propagation profile.

6. Geometric Equivalence of Curvature and Divergence

To demonstrate how the divergence — the local information density $\rho = \partial_\mu \Xi^\mu$ — identically drives this curvature scalar, the displacement field is decomposed into its longitudinal component using a scalar displacement potential χ :

$$\Xi_{;\mu} = \partial_{\mu} \chi$$

Taking the divergence yields $\rho = \square \chi$. Substituting this representation back into the geometric Ricci scalar identity, the term $\square \Xi_{;\alpha}$ becomes $\partial_{\alpha} \rho$ (the spatial gradient of the local density field). The Ricci scalar equation now reads:

$$R = (1/T_0) [\partial_{\mu} \partial_{\nu} \partial_{\alpha} \chi \partial^{\mu} \partial^{\nu} \partial^{\alpha} \chi - \partial_{\alpha} \rho \partial^{\alpha} \rho]$$

Transforming this relation into Fourier space (where $\partial_{\mu} \rightarrow i k_{\mu}$ and $\square \rightarrow -k^2$) allows an analysis of how the density field structures spatial curvature:

$$R = (1/T_0) [(k_{\mu} k_{\nu} k_{\alpha} \chi)^2 - (k_{\alpha} \rho)^2]$$

Since $\rho = -k^2 \chi$, substituting $\chi = -\rho/k^2$ into the first term yields $(k_{\mu} k_{\nu} k_{\alpha} / k^2) \rho$. Squaring this and summing over all indices yields $k^2 \rho^2$, which identically matches the second term:

$$(k_{\alpha} \rho)^2 = k^2 \rho^2$$

This reveals that the curvature scalar R acts as the net dynamic balance of the substrate's local stress gradients:

$$R = (1/T_0) [\partial_{\mu} \partial_{\nu} \Xi^{\alpha} \partial^{\mu} \partial^{\nu} \Xi_{\alpha} - \partial_{\alpha} \rho \partial^{\alpha} \rho]$$

While this algebraic balance yields a net-zero static curvature in idealized, uniform Fourier space, physical gravity manifests dynamically. Real-world localized mass concentrations create localized non-equilibrium propagation delays ($\Delta t \neq 0$), ensuring that localized stress gradients $\partial_{\alpha} \rho$ do not instantaneously cancel out, thereby leaving a measurable macroscopic curvature tensor $R_{\mu\nu}$. When there is a localized concentration of matter, the spatial gradients of this concentration physically compress and distort the adjacent lattice.

7. The Emergent Einstein-Hilbert Action

Substituting this geometric identity directly into the continuous limit of the node potential energy term replaces previous mapping assumptions with an exact physical

correspondence, recovering the continuous Einstein-Hilbert Action with absolute mathematical fidelity:

$$S = \int d^4x \sqrt{-g} [(1/16\pi G) R - L_{\text{matter}}]$$

where the gravitational coupling G_{SI} is derived from the bulk mechanical stiffness of the discrete substrate, corrected for coordinate damping and lattice-viscosity response lags:

$$G_{\text{SI}} = G_{\text{struct}} (L_0^3 / M_0 T_0^2) (1 - f_{\text{visc}}) = 6.6743 \times 10^{-11} \text{ m}^3/\text{kg}\cdot\text{s}^2$$

The baseline structural parameters (L_0 , M_0 , T_0) and the lattice-viscosity correction factor ($f_{\text{visc}} = 0.0008014$) represent the foundational hardware scaling of the medium. For the purposes of this paper, they are treated as fixed, empirically matched constants; their deep mechanical origins within the broader substrate architecture are detailed in separate work.

8. Variational Derivation and Collapse to the Field Equations

To find the equation of motion for the spacetime metric, the continuous action S is varied with respect to the induced inverse metric $g^{\mu\nu}$:

$$\delta S = \int d^4x [(1/16\pi G) \delta(\sqrt{-g} R) - \delta(\sqrt{-g} L_{\text{matter}})] = 0$$

Using standard variational identities for the volume element $\delta(\sqrt{-g}) = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}$ and the Ricci scalar $\delta R = R_{\mu\nu} \delta g^{\mu\nu}$ (where the boundary terms vanish via the metric compatibility theorem), this yields:

$$\delta S = \int d^4x \sqrt{-g} [(1/16\pi G) (R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}) \delta g^{\mu\nu} - \frac{1}{2} T_{\mu\nu} \delta g^{\mu\nu}] = 0$$

where the stress-energy tensor of matter $T_{\mu\nu}$ is defined as the variation of the matter Lagrangian:

$$T_{\mu\nu} \equiv -2 \delta L_{\text{matter}} / \delta g^{\mu\nu} + g_{\mu\nu} L_{\text{matter}}$$

Since the variation $\delta g^{\mu\nu}$ is arbitrary, the terms in the brackets must vanish, collapsing directly into the Einstein Field Equations:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

To obtain the explicit energy-momentum footprint of the substrate, the potential action is varied with respect to the induced inverse metric: $S_V = \int d^4x \sqrt{-g} V(\rho)$. Evaluating the derivative of the substrate potential near the matter-formation boundary, where a structural information tension constant ($\Theta = 187.32$) regularizes the expression, produces the macroscopic output:

$$T_{\mu\nu} \equiv -2 (1/\sqrt{-g}) \delta S_{\text{matter}} / \delta g^{\mu\nu} = \partial_\mu \Xi^\alpha \partial_\nu \Xi_\alpha [K \delta\rho] - g_{\mu\nu} [K/2 (\delta\rho)^2]$$

This matches the right-hand side of Einstein's field equations. The diagonal elements (T_{00}) characterize the static compressive load of standard condensed matter ($\rho > 1$), while the permanent outward residual trace elements (T_{ii}) naturally yield the isotropic displacement pressure observed in the sub-yield regime ($\rho < 1$). The physical interpretation of these regimes as distinct quantum phases (such as dark matter and dark energy) is an emergent feature of the global system and is treated extensively in a companion paper.

9. Background Independence and Translation Invariance

To prove that a global coordinate translation of the displacement field variables does not introduce an unphysical, coordinate-dependent footprint, the Discrete Noether's Theorem is applied directly to the node-update action over the five-dimensional hyper-lattice.

The discrete action of the displacement field Ξ_μ across a set of lattice nodes (n) is defined as:

$$S_{\text{discrete}} = \sum_n [(T_0/2) (\Delta_\nu \Xi_\mu(n))^2 - V(\rho_n)]$$

where Δ_ν represents the forward discrete difference operator between adjacent nodes:

$$\Delta_\nu \Xi_\mu(n) = \Xi_\mu(n + e_\nu) - \Xi_\mu(n)$$

Applying a uniform global translation shift (ε_μ) to the entire displacement field ($\Xi_\mu(n) \rightarrow \Xi_\mu(n) + \varepsilon_\mu$), the shift cancels out identically because the forward difference operator subtracts adjacent node states:

$$\Delta_\nu (\Xi_\mu(n) + \varepsilon_\mu) = \Delta_\nu \Xi_\mu(n)$$

Because the non-linear potential term $V(\rho_n)$ depends only on the local information density ρ_n (the discrete divergence), it is also invariant under global translations ($\delta S_{\text{discrete}} = 0$). Applying the discrete variational principle yields the discrete equation of motion for each node:

$$\sum_n [\Delta_v (T_0 \Delta_v \Xi_\mu(n))] = 0$$

At macroscopic scales far larger than the lattice spacing ($L \gg \lambda$), the discrete difference Δ_v / λ transitions into the continuous partial derivative ∂_v , and the sum converts to a volume integral, recovering the exact covariant conservation of the energy-momentum tensor:

$$\partial_v T^\nu_\mu = 0$$

This proves that macroscopic momentum conservation and background independence are the inevitable geometric consequences of translation invariance operating across the discrete 5D nodes, mathematically preventing the manifestation of an unphysical preferred reference frame.

10. Conclusion

This derivation demonstrates that General Relativity is an emergent behavior. The Einstein-Hilbert action and the Einstein Field Equations are derived as the continuous, large-scale limit of physical tension gradients operating across a discrete 5D information-bearing lattice. By anchoring the coordinate metric to the non-linear potential and displacement gradients of this substrate, the framework provides a concrete physical mechanism for gravitational curvature, background independence, and the conservation of momentum.

While this paper has fulfilled its single, isolated purpose — proving that a 5D strain-tensor field projection reduces identically to standard General Relativity — the boundary thresholds established here interface directly with deeper field-theoretic properties. The sub-yield quantum superposition states ($0 < \eta < 1$), the mechanical origin of probability,

flavor universality deviations, and the number-theoretic derivation of the prime seed scaling constant ($\dagger$) are treated as independent modules and are presented fully in the comprehensive companion manuscript.

References.

0. Fenn, J. (2026). “The Mechaniverse: A Unified Information-Substrate Model of the Universe,” Zenodo, Working Paper. Doi: 10.5281/zenodo.18722388
1. Einstein, A. (1915). Die Feldgleichungen der Gravitation. Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin, 844–847.
2. Hilbert, D. (1915). Die Grundlagen der Physik (Erste Mitteilung). Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse, 395–407.
3. Kaluza, T. (1921). Zum Einheitsproblem der Physik. Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin, 966–972.
4. Klein, O. (1926). Quantentheorie und fünfdimensionale Relativitätstheorie. Zeitschrift für Physik, 37(12), 895–906.
5. Sakharov, A. D. (1968). Vacuum quantum fluctuations in curved space and the theory of gravitation. Soviet Physics Doklady, 12, 1040–1043.
6. Regge, T. (1961). General relativity without coordinates. Il Nuovo Cimento, 19(3), 558–571.
7. 't Hooft, G. (1993). Dimensional reduction in quantum gravity. arXiv:gr-qc/9310026.

8. Verlinde, E. P. (2011). On the origin of gravity and the laws of Newton. *Journal of High Energy Physics*, 2011, 29.

9. Padmanabhan, T. (2010). Thermodynamic aspects of gravity: New insights. *Reports on Progress in Physics*, 73(4), 046901.

10. Wheeler, J. A. (1957). On the nature of quantum geometrodynamics. *Annals of Physics*, 2(6), 604–614.

11. Milne, E. A. (1948). *Kinematical Relativity*. Oxford University Press.

12. Jacobson, T. (1995). Thermodynamics of spacetime: The Einstein equation of state. *Physical Review Letters*, 75(7), 1260–1263.